Reducing Drug Cardiac Toxicity Using Molecular Simulations and Machine Learning

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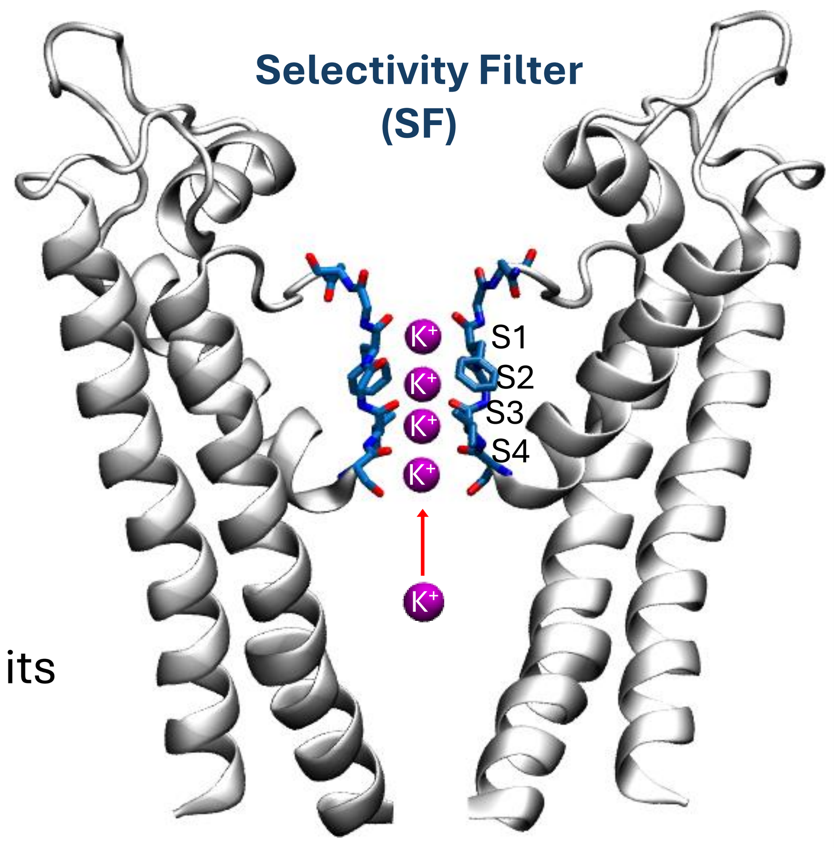
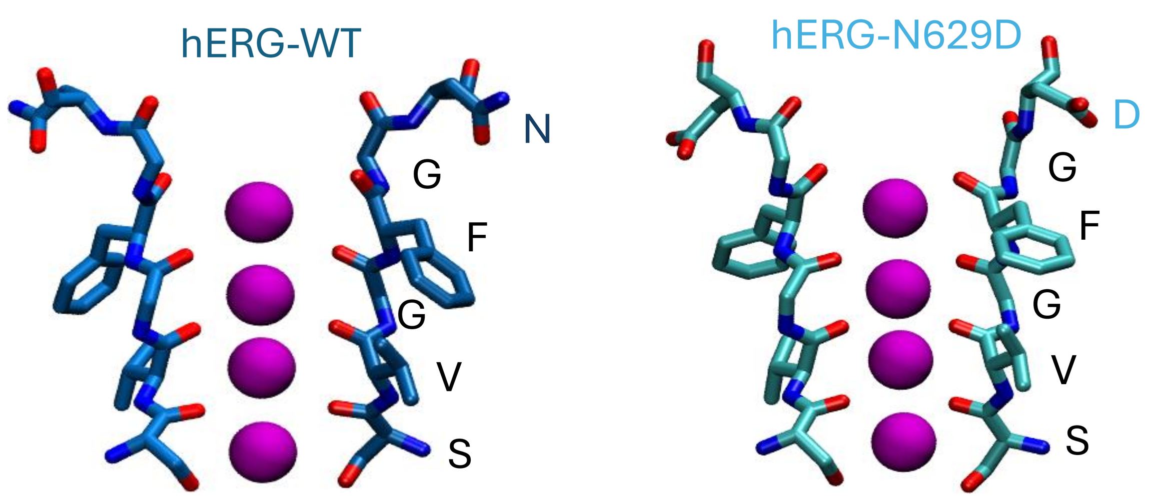
1. Introduction

Ion channels are key drug targets because they regulate essential physiological processes.[1] The hERG channel facilitates the passage of K+ ions and is crucial in cardiac repolarization – helping reset the heart's electrical state after each beat. Many drug candidates fail because they unintentionally block the hERG K+ channel, a major cause of drug-induced cardiotoxicity.[2]



To evaluate inhibitor effects, we first need to understand hERG's native function, including its low conductance and inward rectification,[3] properties that remain elusive despite their importance in regulating heart activity.

Molecular Dynamics (MD) simulations of wild-type (WT) hERG fail to support stable K+ conduction due to force field (FF) limitations, SF instability, and reliance on non-physiological voltages (± 500-750 mV). Recent studies show improved SF stability in the N629D mutant,[4] and newly developed FFs enhance conductance in K+ channels.[5]

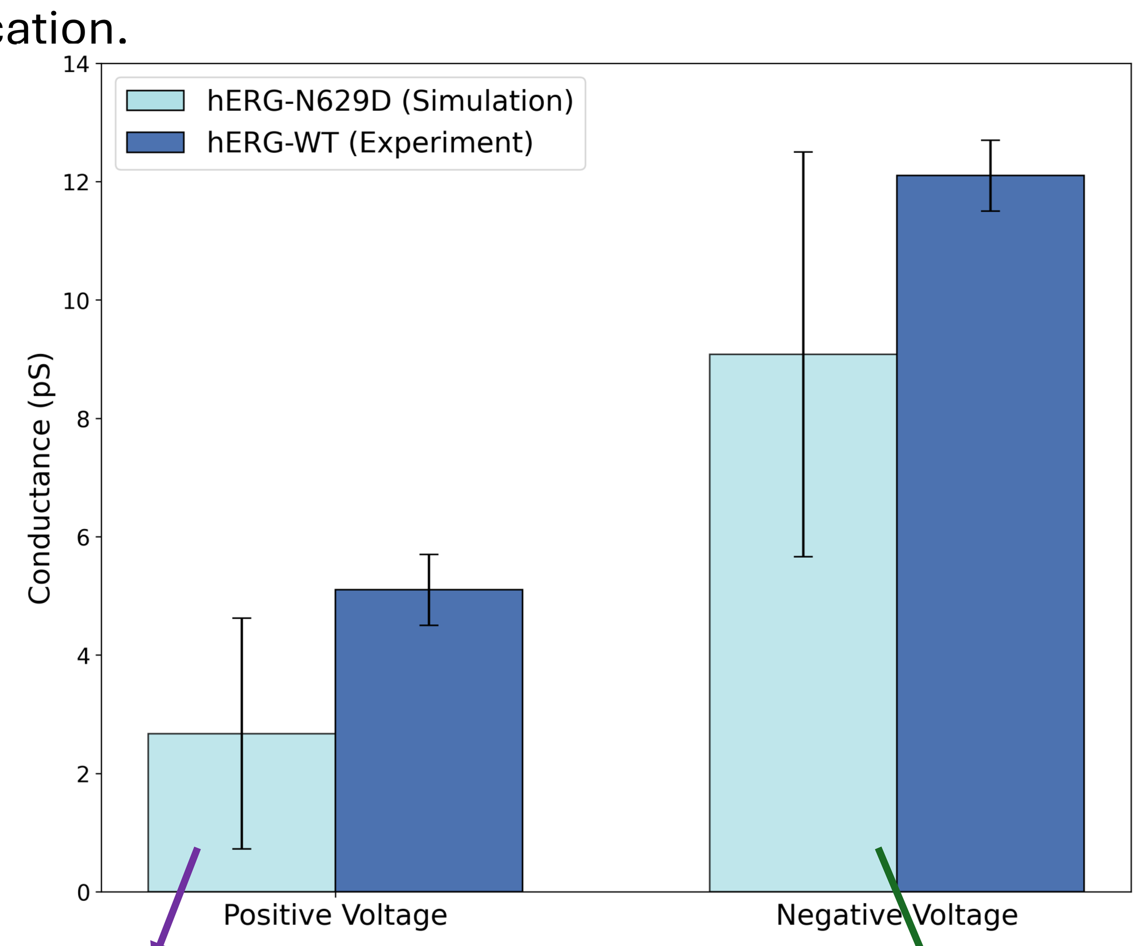


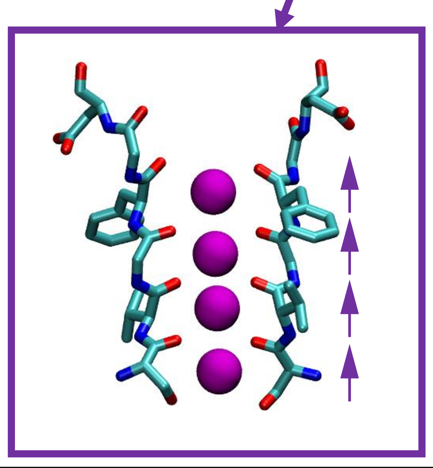
2. Research Aims

- Simulate hERG-WT and N629D systems using improved FFs.
- Achieve stable K+ permeation under near-physiological conditions using moderate voltage (± 300 mV).
- Understand the mechanism underlying inward rectification and low conductance of hERG.

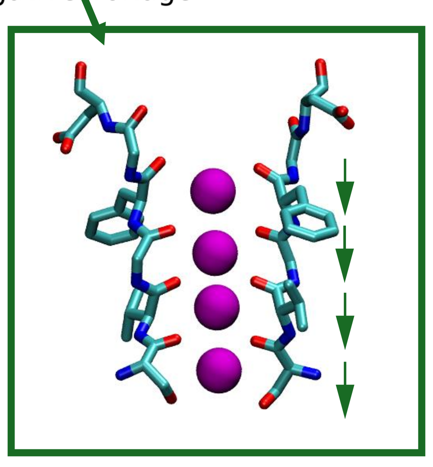
3. Conductance Observed in hERG-N629D with Amber14sb-s3

Systems were simulated using the electronic continuum correction (ECC) optimized Amber14sb FF with 0.78 scaling, including additional scaling of the S3 carbonyl oxygen to enhance conductance.[5] With this setup, the N629D mutant showed conductance within the experimental range for hERG-WT,[6] and preserved inward rectification.



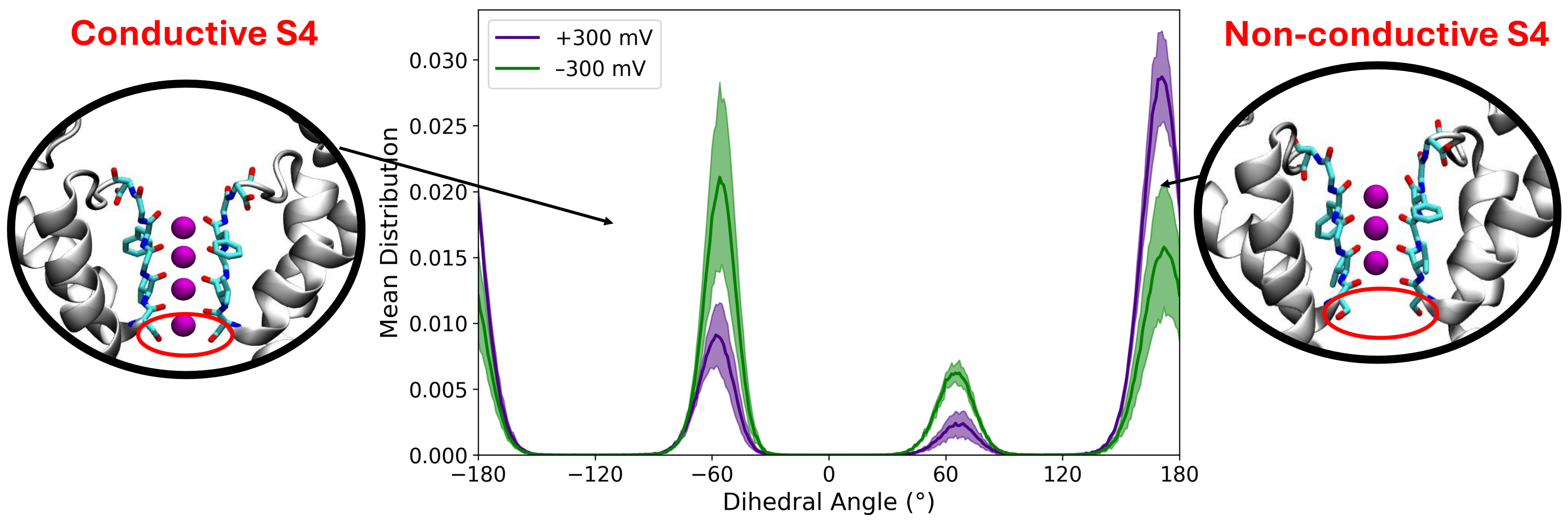


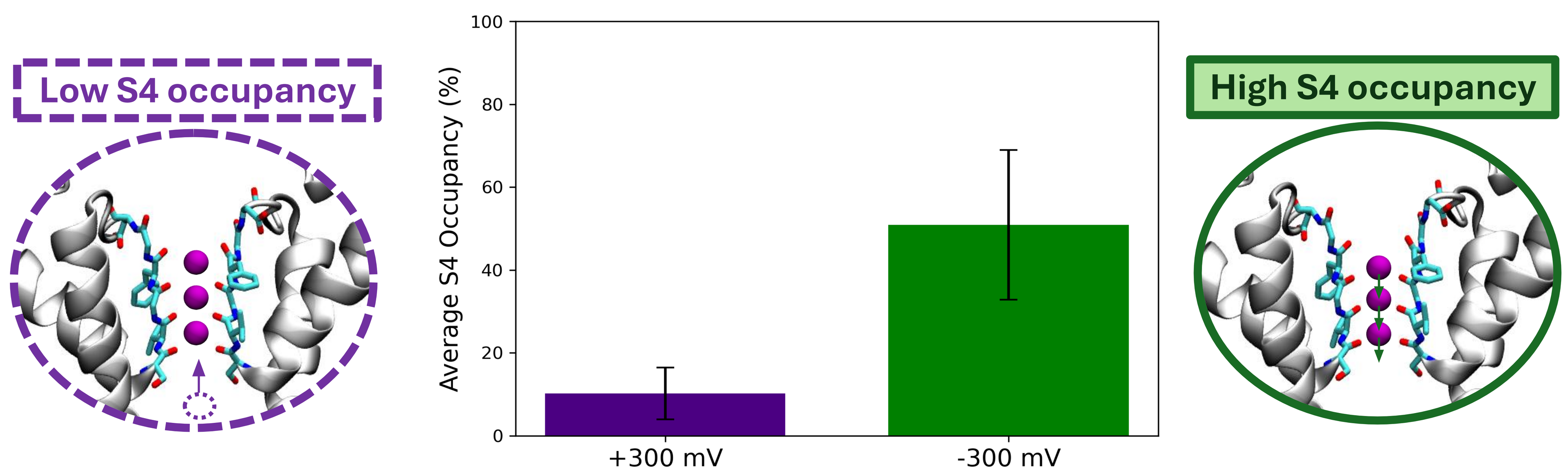
What is causing the inward rectification in hERG?



4. S4 Site as a Basis For Inward Rectification

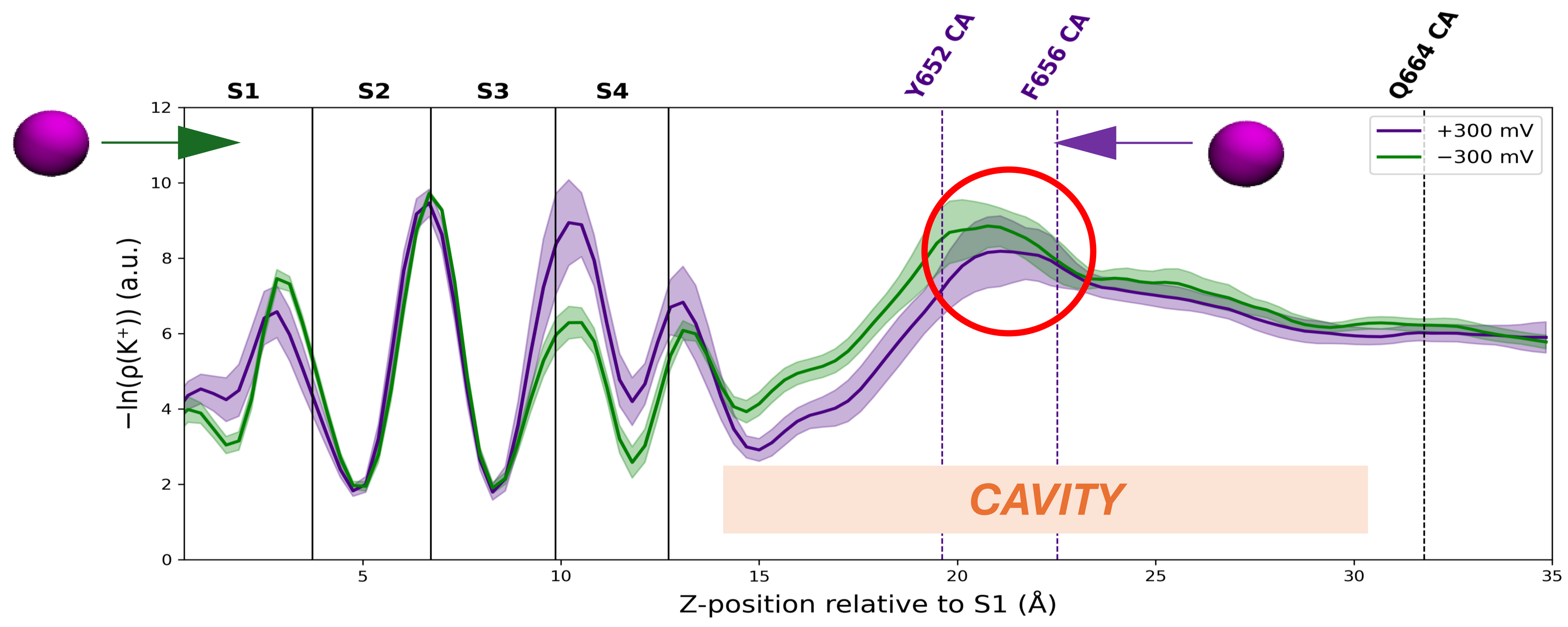
We focused on the serine residue at the S4 site. Its sidechain shifts between conductive (± 60°) and non-conductive (± 180°) states depending on K+ binding to S4. However, S4 occupancy decreases at positive voltage, limiting outward conduction.

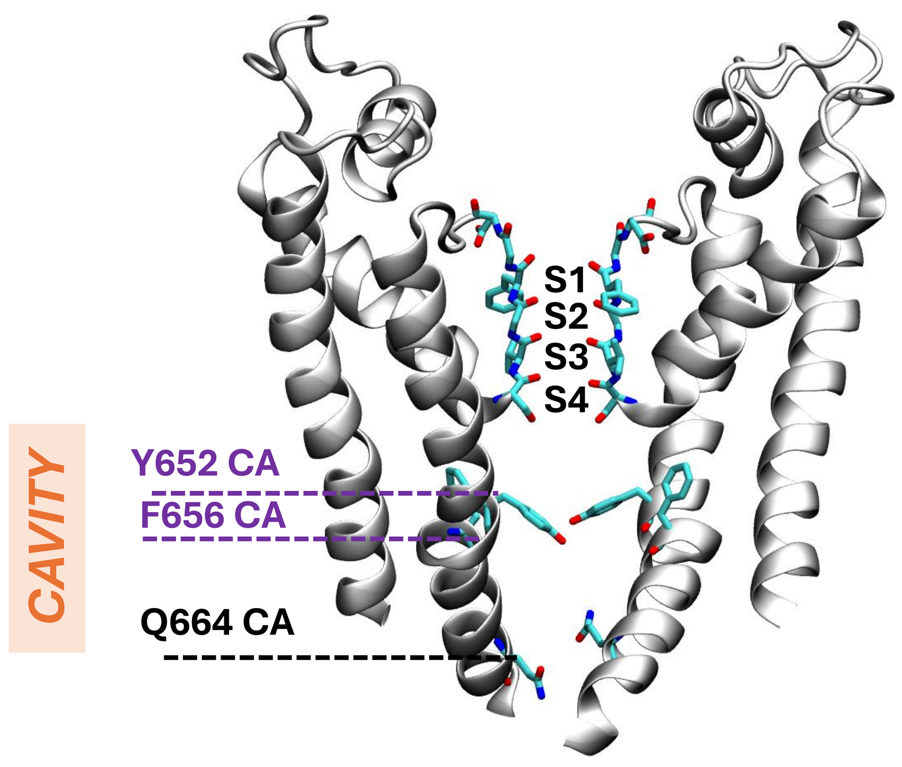




5. Cavity Restricts K+ Entry at Positive Voltages

Mean free energy profiles along the K+ conduction pathway reveal a free energy barrier between Y652 and F656 that limits K+ access to S4 at positive voltage.



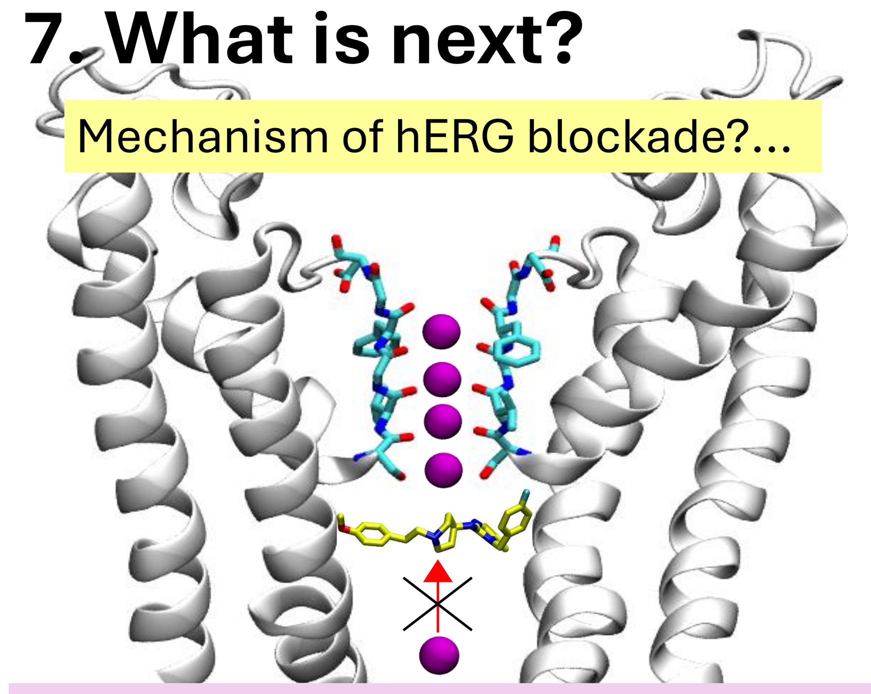


K+ accumulation in the cavity is energetically unfavourable, leading to a reduced S4 occupancy and outward conductance at positive voltage.

6. Conclusions

- Observed K+ permeation under stable, near-physiological conditions using improved FF and moderate voltage.
- Improved FF yield conductance values consistent with experiment.
- Inward rectification in hERG arises from voltage-dependent S4 occupancy, where K+ access from below is impeded at positive voltage, reducing outward conduction.

7. What is next?



7. References

1. G. J. Kaczorowski, O. B. McManus, B. T. Priest and M. L. Garcia, J. Gen. Physiol., **2008**, 131, 399–405.
2. T. M. Creanza, P. Delre, N. Ancona, G. Lentini, M. Saviano and G. F. Mangiatordi, J. Chem. Inf. Model., **2022**, 62, 1170–1181.
3. D. A. Doyle, J. M. Cabral, R. A. Pfuetzner, A. Kuo, J. M. Gulbis, S. L. Cohen, B. T. Chait and R. MacKinnon, *Science*, **1998**, *280*, 69–77.
4. J. Li, R. Shen, B. Reddy, E. Perozo and B. Roux, *Sci. Adv.*, **2021**, *7*, eabd6203.
5. C. Hui, R. de Vries, W. Kopec and B. L. de Groot, *Proc. Natl. Acad. Sci. U.S.A.*, **2025**, *122*, e2423866122.
6. A. Zou, M. E. Curran, M. T. Keating and M. C. Sanguinetti, *Am. J. Physiol. Heart Circ. Physiol.*, **1997**, *272*, H1309–H1314.

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